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Neutrophils can synthesize leukotriene B₄ (LTB₄) by activating the 5-lipoxygenase (5-LO) signaling pathway. LTB₄ is a pro-inflammatory mediator associated with the etiology and progression of atherosclerosis. It can increase function and number of neutrophils in an autocrine manner. Since hypercholesterolemia is associated with an increase in the number and function of neutrophils, we hypothesized that this effect could be mediated through increased production of LTB₄ in neutrophils. Hypercholesterolemia was modeled in Wistar rats by feeding them with a high cholesterol diet. The induction of hypercholesterolemia caused an increase in the plasma levels of LTB₄, following lipopolysaccharide stimulation. This effect was recapitulated in vitro, both in the presence and absence of stimulation with the activator of 5-LO, A23187. Neutrophils in hypercholesterolemia rats expressed similar total levels of 5-LO as control rats, but displayed increased nuclear localization of 5-LO, as well as elevated levels of phosphorylated 5-LO and ERK1/2. In vitro, M β CD/cholesterol complexes enriched cholesterol in neutrophils, resulted in similar changes in 5-LO/LTB₄. In addition, these alterations could be inhibited with the ERK inhibitor PD98059.

Question: Does hypercholesterolemia increase the production of leukotriene B₄ in neutrophils by enhancing the nuclear localization of 5-lipoxygenase?

Answer: Hypercholesterolemia increases LTB₄ production in neutrophils by increasing the nuclear localization of 5-LO, which is the result of its phosphorylation by activated ERK1/2.